INTERNSHIP PRESENTATION

Document OCR & Data Extraction

A web-based digital management system developed for automated document processing, record organization, and data extraction.

PRESENTED BY

Amritanshu Tripathi

ROLL NUMBER & BATCH

25SCS1003005100 | 2CSE10

INSTITUTION & ORGANIZATION

IILM University | Nanoglint Technologies

PROJECT BACKGROUND

Executive Overview

This report highlights the design, development, and practical learnings during an 8-week software engineering internship focused on digital document processing and web management systems.

Organization Profile & Details

HOST ORGANIZATION

Nanoglint Technologies

Nanoglint Technologies is an IT & software-oriented organization offering real-world product exposure. It focuses on practical implementation of computer science concepts, requirement analysis, backend architecture, and production testing environments.

INTERNSHIP DETAILS

Tenure & Scope

Duration: June 18, 2026 – August 18, 2026

Focus Area: End-to-end web system design, document optical character recognition (OCR) workflows, CRUD operations, database schema design, and software verification.

Problem Statement

LEGACY CHALLENGES

Manual Record Keeping

Traditional manual document processing and physical record handling are error-prone, labor-intensive, and inefficient. Information retrieval requires significant effort, and updating records manually leads to duplicate entries and missing data.

PROPOSED SOLUTION

Automated Centralized Platform

A web-based digital platform equipped with OCR capabilities and centralized database storage. The system enables users to capture, convert, search, edit, and safely manage records through an intuitive user interface.

Primary Objectives

- **Automate Data Extraction:** Transition from physical/unstructured document entry to automated digital extraction.
- **Implement Full CRUD Operations:** Enable robust Create, Read, Update, and Delete operations for seamless record lifecycle management.
- **Hands-on SDLC Experience:** Apply complete Software Development Life Cycle principles—from analysis and design to testing and deployment.
- **Enhance Data Retrieval & Accuracy:** Minimize manual data entry errors while providing instant global search capability across digitized records.

Scope of the System

Frontend Interface

Clean, responsive web page structures built for simple navigation, input validation, document uploads, and quick record visualization.

Data Management

Structured database integration enabling real-time record creation, targeted search queries, record updates, and deletion capabilities.

Future Extensions

Planned extensions include role-based access control (RBAC), multi-format export, cloud storage integration, and advanced AI models.

Technology Stack & Tools

Technology / Tool	Category	Purpose & Application
HTML5 & CSS3	Frontend Layer	Structuring application pages and designing modern responsive interfaces.
JavaScript	Client Scripting	Dynamic DOM manipulation, event handling, and client-side validation.
Python & Backend Framework	Application Layer	Executing core business logic, document parsing, and processing user requests.
Relational Database	Data Storage	Persisting structured records, user metadata, and system activity logs safely.
VS Code & Git/GitHub	Development Tools	Code editing, local testing, version control, and codebase tracking.

Three-Tier Architecture

1. Presentation Layer

Built using HTML, CSS, and JavaScript. Renders the graphical user interface, captures document inputs, and handles real-time interaction feedback for end users.

2. Application Layer

Powered by Python backend logic. Processes OCR parsing requests, verifies inputs, executes business rules, and routes calls between UI and storage.

3. Data Layer

The centralized relational storage environment. Responsible for persistent data storage, row execution, and handling CRUD queries reliably.

> Workflow: User → Web Interface → Application Backend → Database

Development Methodology

1. Analysis

Requirement gathering and system feature definition.

2. Design

Wireframing and 3-tier architecture planning.

3. Build

Frontend, backend & DB development.

4. Test & Doc

Bug fixes, functional testing & final report.

Efficiency & Performance Gains

85 %

Reduction in Data Entry Time

Operational Transformation

By transitioning from paper-bound manual entries to an automated OCR-assisted web application, document processing cycles were drastically reduced.

Key outcomes include zero data duplication, instant search retrieval across thousands of indexed records, and significantly lower administrative effort.

System Performance Comparison

The web-based OCR and data extraction system provides substantial improvements in throughput, accuracy, and ease of record management compared to legacy manual processes.

END OF PRESENTATION

Thank You!

Questions & Discussion

Amritanshu Tripathi (Roll No: 25SCS1003005100)

School of Computer Science & Engineering, IILM University

Internship Host: Nanoglint Technologies